AI Malware Investigation Report

Executive Summary

The file `xampp-control.exe` has been analyzed and is classified as Benign by the ML model with a very low malware probability of 2.77% and a risk score of 3/100. Based on static PE analysis, the file is assessed as SAFE / LOW RISK, indicating it does not pose a significant threat to the environment. Further action beyond source verification is generally not required.

Detection Summary

The ML model provided a 'Benign' verdict with a 2.77% malware probability and a low risk score of 3/100. The model's confidence in this classification is 97%. No YARA, CAPA, or Malice AV scan results were provided for further correlation, thus no additional detection indicators were observed.

Final Verdict & Classification

Verdict: SAFE / LOW RISK. This verdict is directly supported by the ML model's classification of 'Benign' and an extremely low risk score of 3/100, indicating no malicious characteristics were identified.

Classification: Benign. The ML model explicitly classified the file as 'Benign' with a very low probability of being malware, and no other detection engines contradicted this finding.

Confidence Assessment: High. The ML model reported a 97% confidence in its 'Benign' classification, which is a strong indicator of reliability for this assessment. The low malware probability further reinforces this confidence.

Technical Findings

The file `xampp-control.exe` (SHA256: ed4d6fdb6248bcff64e5652cd0c9d79c483bace94c1120dc3128645f00a5e5c4) is a Portable Executable (PE) file with a size of 3,368,448 bytes. No specific malicious static PE features such as high entropy sections, suspicious imports, or packed indicators were observed in the provided scan results, aligning with the benign classification.

Behavioral Analysis

Based on the 'Benign' ML verdict and the absence of other detection indicators, the following behavioral aspects are assessed:

* Not Observed: Execution of malicious code or payloads.
* Not Observed: Attempts to establish persistence mechanisms.
* Not Observed: Network communication with command-and-control (C2) servers.
* Not Observed: Unauthorized file system modifications or data exfiltration.
* Not Observed: Process injection, manipulation, or privilege escalation attempts.

Supporting Evidence

['ML Verdict: Benign', 'Malware Probability: 2.77%', 'Risk Level: Low', 'Risk Score: 3/100', 'ML Confidence: 97%', "ML Recommendation: 'The file is likely benign based on static PE features. If the source is trusted, it can be treated as low risk.'"]

Indicators of Compromise

- a01b09b6d27d101391aab54ac0879e5b
- e36b768abc97f755161b0112b01d6644f8db5c60
- ed4d6fdb6248bcff64e5652cd0c9d79c483bace94c1120dc3128645f00a5e5c4
- xampp-control.exe

MITRE ATT&CK Mapping

Risk Assessment

The overall risk level for `xampp-control.exe` is assessed as SAFE / LOW RISK. This assessment is primarily driven by the ML model's strong 'Benign' classification, a very low malware probability (2.77%), and a minimal risk score of 3/100. There is no evidence from the provided scan results to suggest any malicious capabilities or intent, making it highly unlikely to pose a threat.

Potential Impact

Given the ML model's 'Benign' verdict and extremely low risk score, the potential impact of executing this file is assessed as negligible. It is not expected to cause any adverse business or technical impact, data compromise, or system disruption.

Recommended Actions

- Verify the source of the `xampp-control.exe` file to ensure its authenticity and integrity.
- If the source is trusted, the file can be considered safe for deployment and execution within the environment.
- Continue to adhere to standard security practices, including regular system patching and endpoint protection, as a general measure.

Conclusion

Based on the comprehensive static PE analysis performed by the ML model, the file `xampp-control.exe` is confidently assessed as Benign and poses a SAFE / LOW RISK to the environment. The high confidence (97%) in the ML verdict, coupled with the extremely low risk score, provides strong assurance regarding its benign nature.